

Formal Guidelines for Academic Papers

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**Deans' Offices for Business Informatics and
Engineering**

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Foreword

This compendium was created with the intention of providing you, as a student, with guidance regarding the structure of your scientific paper as well as lecturers with a handbook that they can use to determine formal requirements.

In addition to adhering to this set of rules, however, it is of particular importance that you proceed in an evidence-based manner, i.e. systematically and fact-oriented, when writing scientific papers and that you are familiar with the range of methods¹ used in business informatics as well as the research methods used in engineering and apply them in a goal-oriented manner. The selection of suitable research methods has to be derived from the research questions.

In this guideline, you will find, among other things, information on the structure of scientific papers. This information is of an exemplary nature, as the structure of your scientific work is also determined by the research methods you apply.

The primary object of knowledge in business informatics are information systems in business and society. As socio-technical systems, they consist of people, information and communication technology, and organisation as well as the relationships between these.²

In this context, research goals and methods can follow two fundamentally different epistemological paradigms, which are to be understood as the foundation of the methods. The design science paradigm strives to gain knowledge by creating and evaluating IT systems. The behavioural science paradigm, on the other hand, considers the analysis of the behaviour and impact of existing information systems on organisations.³

The epistemic goals of design-oriented research are instructions for the construction and operation of information systems as well as innovations in information systems themselves. Result types of this research include models, methods and solutions, such as prototypes or productive information systems.⁴

Engineering deals with the technical implementation of scientific findings. Primarily, it is about optimal solutions for innovative products or new production processes. Due to digitisation, the boundaries between the disciplines of computer science and engineering are often fluid. There are also many overlaps in the preparation of scientific papers in these disciplines. This guide has been developed against this backdrop. The target group is primarily students from the fields of business informatics and engineering.

¹ Cf. Wilde, T., Hess, T. (2007).

² Cf. Österle, H. et al. (2010), p. 666.

³ Cf. Wilde, T., & Hess, T. (2006), p. 3.

⁴ Cf. Österle et al. (2010), p. 666f.

Research at FOM University of Applied Sciences – with its focus on students who work while gaining their degree – should generate a specific benefit for the economy and society, so we recommend that you pay particular attention to these guidelines.

This handbook has been developed in close cooperation with several professors and is the product of an intense collaboration of all colleagues in the departments of business informatics and engineering. Even though this is not an exhaustive list, particular mention should be made here of the colleagues Prof. Dr. Finke and Prof. Dr. Stern for their extensive draft as well as the colleagues Prof. Dr. Holland, Prof. Dr. Kern, Mr. Mertes, Prof. Dr. Lazar, Prof. Dr. Linssen, Prof. Dr. Münstermann, Mr. Reischl, and Prof. Dr. Willan, to whom the deans owe special thanks for their suggestions. Sincere thanks are given to the department for the support of science and teaching, especially Mr. Limpinsel.

We wish you as students much success in your scientific work.

Essen, November 2021



Prof. Dr. Oliver Koch
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1 Introduction

The preparation of scientific papers usually follows certain conventions that have developed over time. Please note that these conventions have their own peculiarities, depending on the culture and the respective scientific discipline. Furthermore, there have been and still are numerous efforts to standardise these conventions. As a result, many scientific institutions have developed their own standards, leading to a confusing spectrum with a multitude of specifications.

The purpose of convention and standardisation is simplification. A standard should be understood as a sensible recommendation but not as dogma. In this respect, the primary intention of this handbook is to provide guidance. By no means does this handbook claim to be universally valid. It is not meant to define arbitrary rules. Rather, it should be the reasoning behind the recommendations that is convincing. However, every person working scientifically is called upon to think for themselves and to question standards at their own discretion.

The essence of scientific work lies in the comprehensibility and transferability of the procedure and findings. Unlike in mysticism, abilities and successes are not dependent on the acting person and their individual circumstances. Science is committed to objectivity. It claims to be valid independent of place, time and person. The aim of every scientific elaboration must therefore be to be comprehensible and transferable. For this reason, the structure of a scientific paper should not only be based on the ideas of the author but above all on the expectations of the target group.

There is a wealth of literature on the methodology and form of scientific work⁵. The literature contains similar but also divergent views. An overarching consensus regarding the details of writing scientific papers has not been established yet. There are sometimes very different ideas on this subject, especially between the various scientific disciplines. Even internationally, different views can be found on this subject. Therefore, it is recommended to study a range of literature and then follow suggestions that seem reasonable. In addition to well-meaning orientation aids and many examples, this handbook also contains guidelines concerning the preparation and submission of the thesis.

The following sections give recommendations that correspond to widespread conventions. In addition, they also provide examples of typical errors.

To distinguish them, the recommendations are marked with a green tick in the margin.
Examples of typical errors are marked with a red cross in the margin.



⁵ An overview is provided by e.g.: Eco, U. (2010), Klein, A. (2017), Stickel-Wolf, C. (2018) and Theisen, M. (2017).

2 Use

This chapter demonstrates how to use the elements discussed in detail below in a scientific paper. This section, therefore, provides both explanations and guidance.

2.1 Structure of the work

Detailed guidance is provided in section 5. Your scientific paper should contain the following elements if they apply:

- Title page,
- Blocking note if applicable,
- Table of contents,
- List of figures if applicable,
- List of tables if applicable,
- Where appropriate, other lists (list of formulae and symbols, source texts, abbreviations),
- Text part,
 - Introduction,
 - Introduction (What is it about?)
 - Framework, motivation, research gap/practical question.
 - Problem definition (What is to be worked on?),
 - Formulation of research question(s), elaboration, objective of the paper.
 - Approach (How is the problem approached?),
 - Explanation of research methodology (including reasoning behind the choice ("what am I doing and why did I choose this approach")),
 - Course of the investigation (How do I implement the above-mentioned methodology? Summary of the outline in prose).
 - Chapter(s) on the fundamentals (there can be more than one of these chapters),
 - Implementation,
 - One or more chapters, depending on the size of the thesis
 - Leads to the achievement of the objective of the thesis and answers to the research question(s)
 - Final chapter (see subsection 5.1).
 - Conclusion,
 - Outlook ("Further Research" | only if new questions presented themselves in the course of the paper).
- Back matter (see subsection 5.1),

- Bibliography / List of references,
- Appendices, where appropriate,
- Glossary, index if applicable.
- Declaration of academic integrity (see subsection 7.2)

2.2 Language, diction, grammar

- No "I" or "we" phrases,
- Neutral language,
- Do not ask questions (except for research questions),
- Pay attention to grammar including punctuation,
- Adapt quotation marks to the language,
 - German: „...“
 - English: "..."
- Quotation marks are used *only* for direct quotes, not to emphasise terms.

2.3 Illustrations / Source code

Detailed guidance is provided in subsections 5.7 ff.

- Illustrations have a caption (below the image),
- Illustrations are numbered consecutively,
- Illustrations are always referenced in the text,
 - Reference is made before the illustration or on the same page,
 - The reference gives the full name: "Figure 3",
 - References are not abbreviated,
 - Ideally, the reference is integrated into the text:
 - Figure 4 shows ...
 - In the diagram in Figure 2, you can see ...
 - References should be created using bookmarks (for L^AT_EX: `\autoref{label}`) (for Word: cross-reference).
- If a source is used for an illustration, it is cited after the caption in the same way it would be cited in the text. If the illustration is based on a template, the source is referenced as follows: "Adapted from author [citation]".

2.4 Tables

Detailed guidance is provided in subsection 5.6 The formal conventions are similar to those of illustrations.

- Tables have a title (above the table),
- Tables are numbered consecutively,
- Tables are always referenced in the text,
 - Reference is made before using the table or on the same page,
 - The reference gives the full name: "Table 4",
 - References are not abbreviated,
 - Ideally, the reference is integrated into the text:
 - Table 4 contains ...
 - In Table 4, one can see ...

References should be created using bookmarks (in $\text{\LaTeX}$: `\autoref{label}`), (in Word: cross-reference).

- If a source is used for a table, it is cited after the title in the same way as it would be cited in the normal text. If the table is based on a template, the source is referenced as follows: "Adapted from author [citation]".

2.5 Abbreviations

Abbreviations should be used as sparingly as possible. Only abbreviations such as etc., e.g. and those for currencies, weights and measuring units are commonly used. If subject-specific abbreviations are used, e.g. for institutions or laws as well as technical or mathematical abbreviations, etc., then, a list of abbreviations has to be created. Here, automatic generation is preferable to manual generation, too.

Abbreviations should be listed in alphabetical order

- Abbreviations are introduced before their first use. The term is written out in full, followed by the abbreviated spelling in brackets, for example: abbreviation (abbr.),
- A list of abbreviations is not a substitute for their introduction,
- Abbreviations should only be used sparingly and when they are absolutely necessary.

2.6 Technical terms

Technical terms are single words or phrases that have a specific meaning in the respective subject area. For the benefit of precision, the use of technical terms is preferable to paraphrasing. It is also possible to define technical terms that are only valid for the scientific work at hand in order to be able to present certain issues more compactly.

- The spelling of technical terms should be uniform.

2.7 Sources / Quotations

Detailed guidance is provided in section 6.

- The citation style should be consistent. Preferably, established standards should be used, for example:
 - An established referencing style in computer science and engineering is IEEE⁶,
 - Internationally, the Harvard style⁷ (or (cf. Theisen, 2017, p 166)), as well as citations according to the American Psychological Association⁸ are common,
 - You can also use footnotes, as in this handbook.
- Quotes from longer sources or sources with little structure⁹ should be supplemented with a page number or other information that makes them easier to trace,
- Direct quotes should be used sparingly,
- If a source is used for a longer section of text, this should be clearly communicated to the reader *in advance*; for example: The following passage is a summary of Gutenberg's work [citation].

⁶ <https://thesius.de/blog/articles/zitieren-ingenieur-ieee-din-iso-690/>.

⁷ <https://usingsources.fas.harvard.edu>.

⁸ <https://www.apastyle.org>.

⁹ Primarily consisting of continuous text

3 Stylistics

The commitment of a scientific work to general validity is supported by a number of stylistic devices. Since the work is intended to convey knowledge independent of a person, place, and time, you should refrain from using the first person singular. Therefore, you should avoid phrases such as the following:

- I find the *Internet of Things* very exciting and therefore would like to elaborate on it,
- Because searching for literature is very long and tedious, I like to use Wikipedia,
- My supervisor gave me the advice to add a lot of comments to my source code. I have followed this advice,
- Social media are great for interpersonal engagement. That is why I unreservedly recommend them for internal company communication as well.



If you really want to communicate something personal in a scientific paper, for example your personal motivation, an acknowledgement, or personal background information, you can do so in a preface. The preface is placed before the table of contents and is not given a section number. It is not listed in the table of contents and does not contain any information that is essential for understanding the actual paper.

For the work to be timelessly valid, it is advisable to use the present tense. Whenever someone reads the text, it is current and, thus, always valid. For this reason, you should only make use of the past tense for actual historical facts. In general, the use of past tense can be a warning signal, because it often occurs when the author is rambling. Note the difference between an analytical line of argument, in which thoughts build on each other logically, and a chronicle, in which the course of events is presented in chronological order.

A scientific work should not take itself as the object of observation. This carries the risk of reflexivity, which can lead to logical inconsistencies. The paradox of the Cretan Epimenides illustrates this well. His statement is: "All Cretans always lie". If the statement is true, it also applies to Epimenides himself, which is why it must be false. In logic, then, the truth value of a self-reference can be indeterminate, which is why it should be avoided.

The following statements are therefore rather improper:

- This work aims to determine the energy consumption of the internet,
- This work demonstrated that cloud computing is a viable option for small and medium-sized enterprises.



Such statements can easily be re-phrased without self-reference to the work.

- The aim is to determine the energy consumption of the internet,
- Cloud computing is a viable option for small and medium-sized businesses.



Commitment to validity is also demonstrated through obvious correctness. In addition to correctness of content, this commitment can also be expressed by observing the rules of spelling, grammar and punctuation. When reading a paper riddled with errors, the reader will easily start to doubt the correctness of its facts and conclusions.

A scientific paper should primarily focus on one line of argument. For this reason, precise terminology is important. Especially for terms that are used in different ways, a precise definition is recommended. The use of definitions developed by the author based on other definitions or combinations thereof is permissible. It is also permissible to introduce special terms, however, those terms should be defined as precisely as possible because the readability of a text is generally improved by the use of distinct terms and precise language. On the other hand, frequent abbreviations, imprecise terms and linguistic redundancies such as the frequent use of filler words have an adverse effect.

4 Methodology

A scientific elaboration is generally based on the application of one or more approaches to gaining knowledge. Typical methods are, for example:

Research: This method essentially uses literature, i.e. books or journals. In addition, other materials can be used, such as conference papers, company publications or documentations. Research is a method that is to be used for all scientific work.

Survey: A common means of obtaining data are surveys. There are different types, for example a poll that addresses a target group that is as representative of the population as possible. This is to be distinguished from the expert interview, which aims to gather specific knowledge. Different qualitative and quantitative methods can be used to evaluate the survey material. There is a good selection of literature on these methods that should be taken into account, for example the evaluation of qualitative methods according to Mayring¹⁰.

Experiment: The classic scientific approach to gaining knowledge is the experiment. It can be a laboratory experiment or field experiment, for example. The subject of a paper that uses an experiment is the description of the design and execution of the experiment as well as the compilation and interpretation of the results.

Design: A constructive approach to developing solutions is the design approach. Different concepts exist to systematise this approach methodically, for example:

- Design Patterns,
- Design science research,
- Design thinking,
- Morphological analysis.

Construction: Especially in engineering, construction is a method for solving a usually technical task.

Programming: Various problems can be solved by suitable software. Their creation can also be the subject of the approach. The actual source code is less important than the algorithms, data structures, interfaces and interchange formats. Special attention has to be paid to their documentation.

These guidelines primarily contain general information on scientific work and rules for formatting a seminar paper or thesis at FOM University of Applied Sciences. They do not cover research methods in detail.

¹⁰ Cf. Mayring, P. (2015).

However, please note that when planning an academic paper, it is essential to choose the research methodology and approach carefully. State your choice explicitly and give reasons for it. Generally, the introduction is a good place to do so.



5 Structure and layout

For scientific papers, the structure is of central importance. Its primary purpose is to present a complex subject matter in a clear and comprehensible manner. In scientific literature, there are a number of useful and established conventions; they are presented below.

5.1 Structure

A scientific paper devoted to a specific topic generally has the following structure:

- Front matter,
- Introduction,
- Fundamentals,
- Implementation,
- Conclusion,
- Back matter

This outline represents a functional approach, not a thematic one. The first instinct when creating an outline is often to structure it with respect to aspects of its content. It then frequently turns out that each of these aspects have a fundamental and an implementation component. Thus, if the thematic structure is implemented consistently, there are usually mental leaps between fundamental principles and implementation. This confusion can usually be resolved by using the pattern mentioned above.

The purposes of each section are as follows:

Front matter: The front matter begins with the title page. There are no strict rules regarding the layout but the following information must be included:

- Identification (name and matriculation number) of the author; the name is often set in the second largest font size¹¹,
- Title of the work, usually set in the largest font size,
- Type of thesis (Bachelor's thesis or Master's thesis),
- University and study programme,
- Name of the supervisor,
- Date and place of completion.

Appendix I shows a sample title page for theses.

If applicable, a blocking notice follows immediately after the title page, see also section 7.3.

¹¹ Please do not include address or contact details.

If applicable, the preface appears before the table of contents. It does not have a chapter number and is not listed in the table of contents.

Various lists are part of the actual work and appear in the front matter. They are to be included as needed if they serve a purpose:

- Table of contents (the table of contents itself does not appear in the table of contents),
- List of figures,
- List of tables,
- List of abbreviations; specific abbreviations should be listed here but not common ones that are universally accepted (e.g., i.e., etc.) – these should generally be avoided or only used sparingly;
- Glossary – this list should compile and explain essential terms. It may be useful to make a reference to the first appearance of the term in the text;
- List of mathematical symbols and indices; this list is useful if mathematical formulae are used to a significant extent, see also section 5.8;
- Directory of source code; if software excerpts or configuration data is referenced, it may be useful to include them in this directory.

The listed indexes do not receive section numbers, but they are listed in the table of contents.

The pages of the front matter are generally numbered in Roman numerals. Thereafter, the page numbering is changed to Arabic, starting with one.

Introduction: The introduction should lead to the topic. To this end, it contains several sections, which can be numbered or structured as simple paragraphs, depending on the scope of the introduction. First, the problem and objective as well as the motivation of the scientific work should be presented.

The introduction is a suitable place to formulate central questions, which the paper will aim to answer. In general, it can be useful to gather relevant questions, which would be asked by an intended readership, through methods like brainstorming in the initial stage of the work. Structuring these questions and tracing them to central issues then leads to the formulation of research questions.

Then, the introduction explains and gives reasons for the approach adopted to address the research questions.

It can be useful to establish conventions for the work. These can be design conventions such as the use of certain fonts for certain purposes. Furthermore, boundary conditions that depend, for example, on location or time can be specified for the work.

Fundamentals: It makes sense to build a scientific paper on already existing findings and related studies. For reasons of comprehensibility, these findings are to be

summarised. The chapter on fundamentals presents the relevant fundamental principles as the result your research. Please refrain from reproducing generally known facts or essentials that are not relevant for the rest of your scientific paper.

Fundamentals from different subject areas should appear in separate subsections.

Chapters covering fundamentals can and should be given descriptive titles – they do not have to be titled Fundamentals.

The chapter(s) on fundamentals cover only topics and terminology that are relevant for the paper and necessary to approach and answer the research question.



Implementation: A scientific paper should provide answers to questions through the systematic application of suitable methods. Answering the questions is the subject of the implementation part of the scientific work. This part provides space for creative solutions, which may consist of clear results, assessments or recommendations.

Conclusion: The final chapter always contains a summary of the main results. These are presented in a concise and succinct manner in the form of a conclusion. Under no circumstances should the final chapter contain a review of the work or reflections on the work itself.

The concluding chapter may give an outlook if, in the course of the work, significant new questions for future consideration have presented themselves. An outlook is unsuitable for discussing subtasks of the thesis that have not been addressed within the paper.

Back matter: Includes at least the list of references. See section **Fehler! Verweisquelle konnte nicht gefunden werden.** on how to create this list. As with the other indexes, the list of references does not have a section number, however, it is listed in the table of contents.

A keyword index is particularly useful for finding important terms. Such an index can be included after the list of references. It can be useful to create a keyword index in the course of quality assurance (see section 7.5). Review the text for important terms while proofreading its final version and include these terms in a keyword index. It is highly recommended to use the respective feature of the text-processing programme to create the keyword index and identify all the references to the key terms within the academic paper.

An appendix is also optional. It can include material that is not immediately necessary for understanding the text, but which gives the reader the opportunity to learn more on how insight was gained. This includes, among other things:

- Transcripts of interviews,
- Source code,

- Product specifications,
- Technical drawings and plans.

Where appropriate, the appendix is to be divided into subsections. The numbering in the appendix is in capitalised Roman numerals. The appendix and, where appropriate, its subsections are listed in the table of contents.

Particularly extensive or less relevant material is not reproduced in the appendix, but is stored on a data storage device.

The figure below summarises common errors related to the proportions of the structure of a scientific paper. The segments refer to the sections described above:

1. Introduction
2. Fundamentals
3. Implementation
4. Conclusion/outlook

right:



wrong:



Note: The sections may consist of several chapters.

5.2 Text layout

5.2.1 Page layout

In general, a scientific paper is submitted in DIN A4 format. This influences various measurements, which are listed below; they are to be understood as rough guidelines that can be deviated from within certain limits. On each page, space should be provided for

Header: Section headings and page numbers may appear here. On the first page of a chapter, there is usually no header.

Body: The actual text.

Footer: Footnotes and, if applicable, page numbers appear here.

These areas have the same width of about 140 mm. The following values provide guidance for the distance of the text from the page margin:

- Top: 4 cm (plus a distance from the side edge of the header: 2 cm),
- Bottom: 2 cm (as well as a distance from the side edge of the footer: 2 cm),
- Left: 4 cm,
- Right: 2 cm.

For duplex printed documents, odd pages are on the right and even pages on the left. Page numbers should be arranged symmetrically, e.g. on the outer edge of each page. The gutter margin should be the same size on the left and right pages so that the type areas on the front and back of a sheet of paper overlap as closely as possible.

5.2.2 Font

The same font must be used throughout the paper.

- **Guidelines for font size and line spacing:**
 - Line spacing in the text: 1.5 lines, font size 12 pt (Times New Roman). Since there are many different fonts, the choice of the font size should be adapted (e.g. Arial 11 pt) so that a page consists of about 37 lines with about 60 characters each or 2,200 characters per page,
 - The spacing for headings including the captions of figures, tables and formulae should be 12 pt before a paragraph and 6 pt after a paragraph,
 - Footnotes use single line spacing, the font size is 10 pt (Times New Roman) or 9.5 pt (Arial).
- **Use italics, boldface and underlining within the text to emphasise certain statements only as an exception and in consultation with the first examiner. Proper names can be set in italics or in capital letters. Underlining is usually not used in professional typesetting.**

In general, highlighting and effects should be used sparingly and consistently.

5.2.3 Headings

Each chapter, section and subsection has a heading. As a rule, headings are set in bold and in a larger font size than the main text. The font size decreases with each level.

- **The line spacing for headings including the captions of figures, tables and formulae should be 12 pt before a paragraph and 6 pt after a paragraph,**
- **For the outline of a thesis, the decimal outline is recommended:**

3
3.1
3.1.1
3.1.1.1
3.1.1.2
3.1.2
3.2
4

- Numbering is not required more than four levels deep.

The granularity of the work in terms of outline depth should also be chosen as consistently as possible.

Only the numbered headings are included in the table of contents. In general, a page should not be listed more than once in the table of contents.

Inserted texts (e.g. chap. 3 text chap. 3.1 text) between chapters of different outline levels only serve the purpose of guiding the reader and should be used as consistently as possible if they are used at all.

If a level of the outline is sub-divided – for example to create subsection 3.3.1 – it should have at least one other subsection of the same level (namely 3.3.2, etc). There should be no text that cannot be attributed to a level of the outline.

5.3 Paragraph

Paragraphs are used to structure the text.

- The basic setting for spacing is 0 pt before a paragraph of text and 6 pt after a paragraph,
- The typographic alignment for all paragraphs should be justification.

Use hyphenation so that the spaces between words do not become disproportionately large. However, if possible, no more than three lines in a row should end in hyphenation.

Generally, page breaks occur within paragraphs. Avoid single lines at the end or beginning of the page.

5.4 Footnotes

Footnotes are used to provide additional details and information on the issue at hand. These can be general comments or references to sources (for details see section 6.1).

Footnotes are separated from the main text at the bottom of the page by a subtle line. Footnotes are numbered consecutively with Arabic numerals. The font size of the footnote is slightly smaller than that of the main text.

If footnotes are used in connection with tables or figures, then it is often useful to include the footnotes directly below the table or figure. In this case, a separate numbering space and system can be applied, using the Latin alphabet, for example. Table 1

5.5 Lists

It is often useful to separate information that can be structured easily from the continuous text and to prepare it separately. Various structuring techniques can be used for this purpose:

- Bare outlines,
- Description lists,
- Bulleted lists.

Bare outlines are useful when the order of the bullet points is irrelevant and the bullet points cannot be summarised by concise terms. Each bullet point is marked by a prominent symbol at the beginning. This can be a dash, a bold point or another symbol.

The individual items in the outline end in punctuation marks. As a rule, there is a full stop at the end of the last item. Alternatively, punctuation marks can be omitted at the end of each item. The outline layout should be consistent throughout the entire document.

Bare outlines can be nested. For nested outlines, each outline level has its own marking symbol. These symbols are to be used consistently throughout the entire work.

If the text of an outline item is longer than one line, the typographical alignment for this text should be justification.

It is a common mistake to mention the number of bullet points when referring to bare outlines. The following fictitious example demonstrates this.



The following three reasons support the implementation of the recommended measures:

- Technology,
- Cost efficiency,
- Organization,
- Law.

Here another item may have occurred to the author at a later point in time. However, with the addition of this item, the reference to three reasons becomes false. Therefore, the number should simply remain unnamed. If the reader really wants to know how many reasons there are, they may count them.

Description lists can be structured in the same way as bare outlines. At the beginning of each bullet point, however, there is not a prominent symbol but a highlighted term which is then explained. An example of this can be found in section 5.1.

If a meaningful hierarchy or order can be established for the items on the list, then they should be presented in a list with numerical or alphanumerical prefixes. The numbering can be in Arabic or Roman numerals, or the Latin alphabet. In the case of nesting, the numbering of the different levels should be coherent throughout the list.

5.6 Tables

Tables are used to present information that is uniform to a certain degree in a concise manner. Typical examples are currency exchange rates (see Table 1) or natural constants (see Table 2).

Tables have titles and are numbered consecutively. These titles and numbers are recorded in the list of tables. The title appears above the table.

Table 1: Euro exchange rates of currencies

Currency	Abbreviation	Exchange rate*
US dollars	USD	1,13
GBP	GBP	0,85
Swiss franc	CHF	1,12
Japan Yen	JPY	125,23
Russian rouble	RUB	73,01

*As of: 2019-03-21

Table 2: Natural constant

Constant	Abbreviation	Value	Unit
Speed of light	c	299 792 458	m/s
Gravitational constant	G	$6.674\,08 \cdot 10^{-11}$	$\text{m}^3/\text{kg s}^2$
Avogadro constant	N_A	$6.022\,141 \cdot 10^{23}$	1/mol
Boltzmann constant	k_B	$1.380\,648 \cdot 10^{-23}$	J/K
Electric constant	ϵ_0	$8.854\,188 \cdot 10^{-12}$	F/m
Vacuum permeability	μ_0	$1.256\,637 \cdot 10^{-6}$	H/m

There are many common errors regarding the design of tables. Table 3 shows several of these errors:



- Too many lines – lines should be used sparingly and only when they are necessary to comprehend the table;
- Information that appears in each row of a column should be specified in the column header, for example, units;
- The grey background of the first line makes it stand out but also makes it much harder to read; colours should be used sparingly and only if clarity cannot be achieved in any other way;
- The data in the table has an unrealistic number of decimal places – data should only be reported with an accuracy that is known with certainty.

Table 3: Runtimes

Test no.	Runtime system A	Runtime system B
Test 1	1898.057460109279 min	3469.055951970739 min
Test 2	1733.594568021676 min	898.314131902552 min
Test 3	3059.658113880700 min	3372.656940671686 min
Test 4	1010.131501996570 min	2471.480081970486 min
Test 5	741.528261512464 min	1264.966288805787 min
Test 6	1927.7010182982 min	899.160852019618 min

Table 4 is an example for a better table design. However, it is often even better to replace a table with a diagram. Figure 1 contains the contents of Table 4 graphically.



Table 4: Runtimes system comparison

Test no.	A runtime/min	B runtime/min
1	1898.05	3469.05
2	1733.59	898.31
3	3059.65	3372.65
4	1010.13	2471.48
5	741.52	1264.96
6	1927.70	899.16

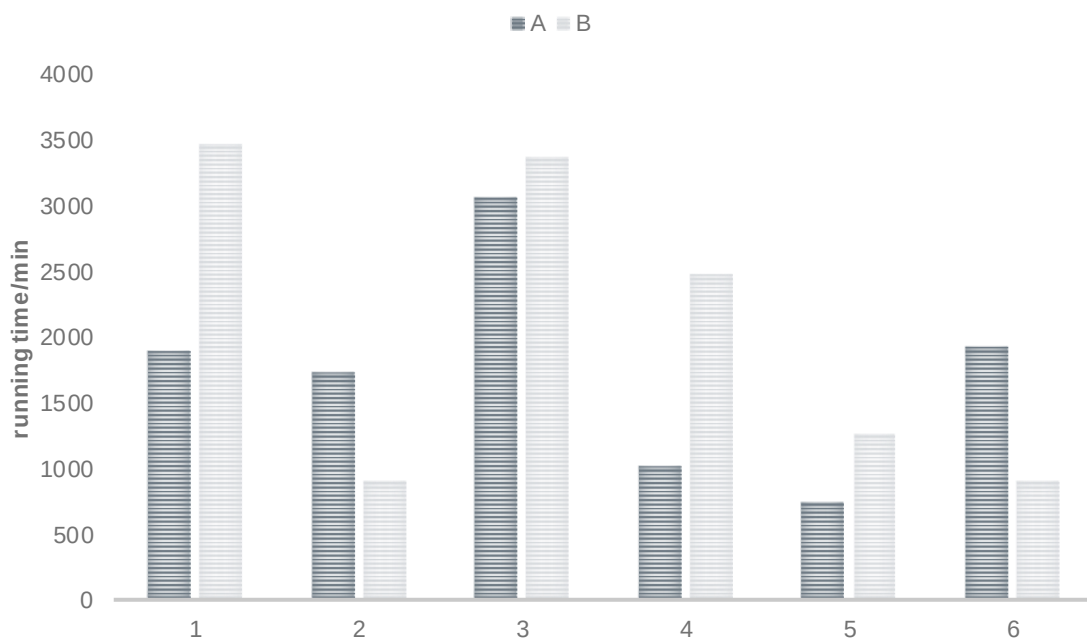


Figure 1: Runtime system comparison

Another common misuse of tables is the listing of comparisons, for example the textual comparison of advantages and disadvantages. This should only be done in the form of a table if the texts are short and if each advantage and disadvantage is associated with a similar aspect. It is better to present such comparisons as lists of descriptions or outlines.

Tables are used for clarity. They should therefore not extend over several pages and should also not be rotated to landscape format. If a table actually extends over more than one page, the table title and the header have to be repeated on the following pages.

Before you start designing an elaborate table, check if other formats are more appropriate.

5.7 Illustrations

A scientific text benefits from the use of illustrations. Design aspects as well as technical aspects have to be taken into account. In terms of design, there is a distinction between schematic representations and realistic illustrations. The former are, for example, schematics, diagrams or plans. The latter include photos and scans. It is recommended to use special programs such as Visio, Inkscape, Gnuplot, R or spreadsheet software to create schematics and diagrams. If the design of an illustration can be influenced, the following recommendations should be observed:

- Less is more – remove all elements of an illustration that are meaningless. This applies, for example, to
 - Unnecessary colours and gradients,
 - Shadows and other 3D effects,
 - Funny characters.
- If colours are used, their meaning should be recognisable. Colours can contribute to clarity, especially in plans that are more complex. Colours have proven very useful for electrical circuit diagrams, for example,
- Reduce text to the absolute minimum in illustrations. A sans serif font is recommended. The font must be large enough to be legible,
- An illustration should show the most essential aspects. Often, illustrations from outside sources are used that only approximate the purpose of the illustration. In these cases, consider using the existing image as inspiration for the design of your own illustration, see for example Figure 2: Design sketch for a prototype¹²,
- Illustrations do not have a frame.



¹² Cf. Spiller, F., (2019).

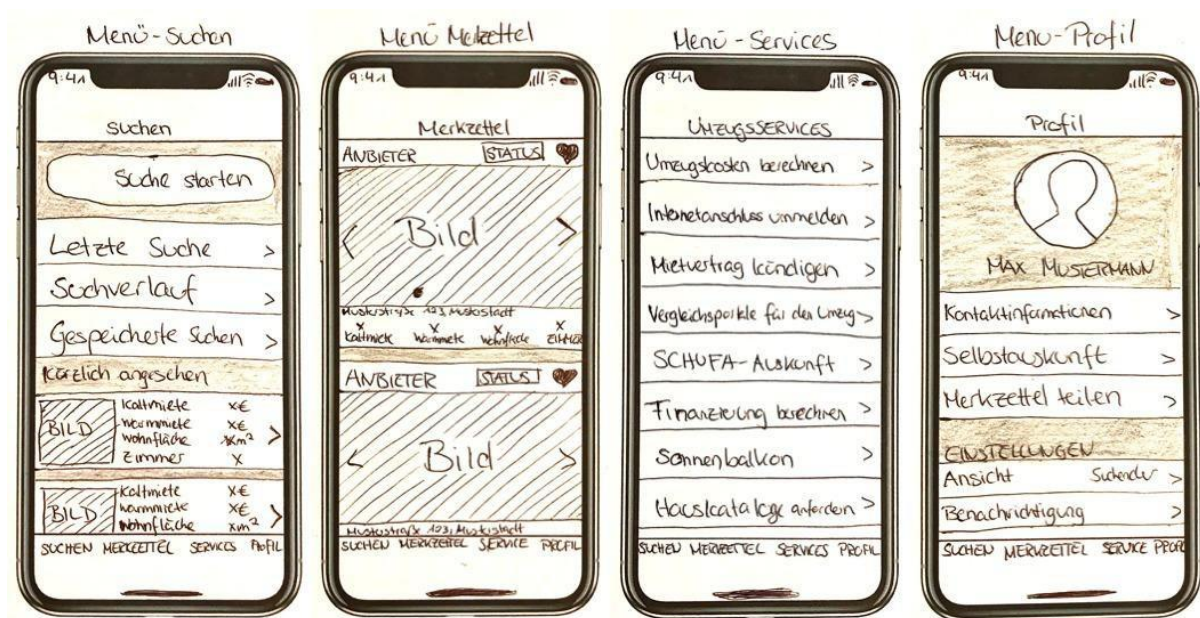
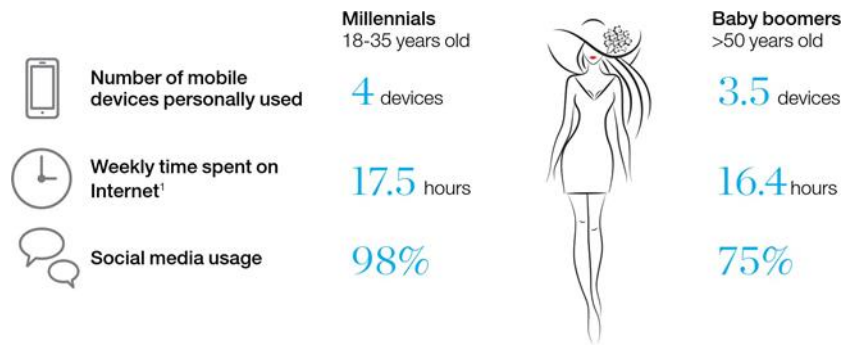


Figure 2: Design sketch for a prototype

Figure 3, Figure 4, Figure 5 and Figure 6 are examples of suboptimal illustrations. Aspects that need to be improved are:

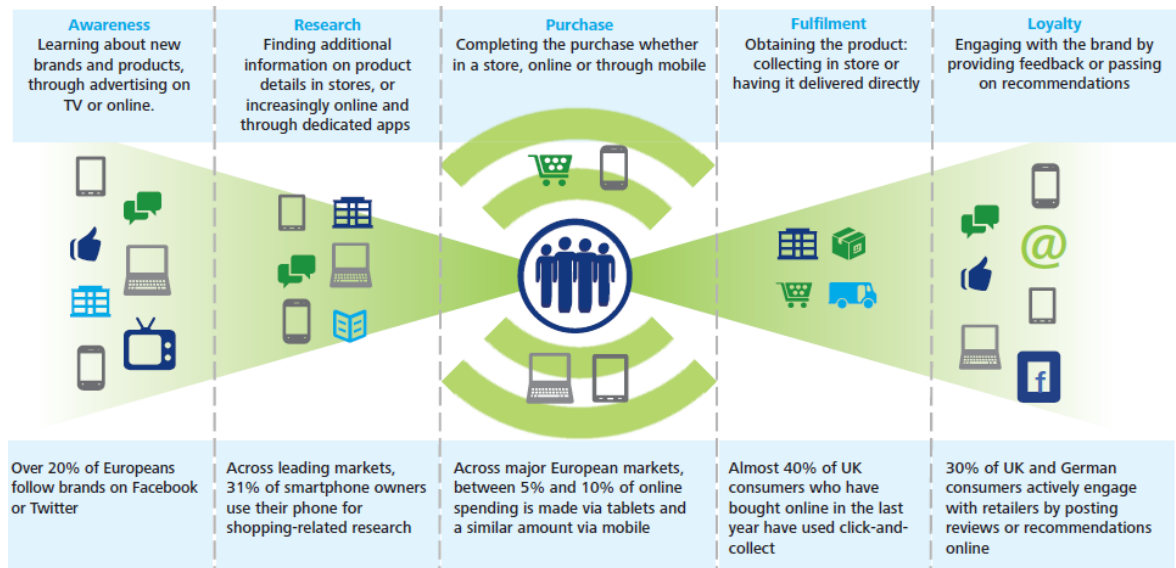


- Representation of essential statements in terms of numbers instead of by size of graphical elements,
- Distracting and meaningless design elements (Figure 3),
- Too much text and poor readability (Figure 4),
- Too small font and pointless design elements (here the green details) as well as blue font on a blue background (Figure 5),
- Symbols that are meaningless or difficult to interpret, as in Figure 3, Figure 4 and Figure 5,
- Figure 6 has too much text, the density of the texts within the columns as well as the serif typeface have are suboptimal. A description list would have been more appropriate.

Figure 3: Comparative representation¹³

5 typical Luxury brand personae				
MILLENNIAL	BUSINESS TRAVELLER	TOURIST	DEAL HUNTER	VIP
<i>"I want to buy everything, anytime, anywhere"</i>	<i>"I want an easy, convenient and fast shopping experience"</i>	<i>"I like to visit stores abroad & enjoy local Luxury experience"</i>	<i>"I want to buy my products at the best price available"</i>	<i>"I like to feel that I am a unique customer to my preferred brands"</i>
22-37 years	30 - 60 years	No specific age	No specific age	> 40 years (mostly)
- Shops preferably on mobile app - High sensitivity to brands with strong presence on social media - Seeks more an experience than a product - Expects seamless and integrated omnichannel services	- Shops for close family (wife / husband, kids) preferably in duty free & department stores - Has little time and needs flexibility - Seeks efficiency & convenience by leveraging commute time	- Seeks brand experience in brand and department stores - Shops specific local assortment / exclusivities for him/herself and relatives - Looks for Detax Hedonistic & open minded, inclined to co-creation	- Seeks for price more than product and experience - Knows what s/he wants - Benchmarks prices from all eCommerce physical & digital channels	- Looks for sales associate engagement and has preferred high-end brands - Seeks dialogue with brand experts - Expects premium & personalized service along the customer journey

Figure 4: Buyer types¹⁴¹³ Cf. McKinsey&Company, (2018).¹⁴ Cf. PWC, (2018).

Figure 5: Omnichannel ¹⁵

Investitionsprojekte	Organisationsprojekte	F & E-Projekte	Komplexe Dienstleistungsprojekte
- Großanlagenbau (Stahl-, Kohlekraftwerke) - Einrichtung einer neuen Fertigungslinie - Kauf/Verkauf eines Unternehmens (M&A)	- Einführung einer neuen Struktur (Change Management) - Einführung eines Planungskonzeptes - Marketingprojekte - Organisation einer Messeveranstaltung	- Entwicklung einer neuen Maschine, eines neuen Medikaments, einer neuen Software oder eines neuen Automobils - Entwicklung neuer Herstellungsverfahren	- Verlegung eines Braunkohlebagger - groß angelegte Instandhaltungsmaßnahmen, z. B. Generalüberholung eines Kraftwerks

Figure 6: Application areas of project management ¹⁶

¹⁵ Cf. Deloitte, (2014).

¹⁶ Bergmann (2016), p. 232.

As a rule:

- Illustrations are provided with a title and sequential numbering with which they are included in the list of figures,
- The caption of an illustration is set **below the illustration**, in contrast to other objects such as tables and lists.

An illustration should be scaled so that it can be easily recognised. Similar illustrations should be presented at a similar scale and should not be abused to meet a page limit. Also, the page limit should not be the reason for relocating an illustration to the appendix.

As a rule, illustrations are centred, alternatively, text can wrap around them.

Illustrations should appear close to the text passages they explain. If many illustrations are used, this is not always possible. Therefore, it is important to consistently title and number illustrations appropriately.

5.8 Formulae

Many facts can be elegantly described by mathematical formulae. Typesetting conventions for mathematical formulae have been established and should be observed. Essential rules are:

- In a formula, abbreviations are used to indicate quantities. These abbreviations are introduced before the first use in the text. If formula symbols are used widely within the paper, they should be listed in a separate directory.
- Variable identifiers are set in italics. Identifiers of constants and operators are set upright, see equations 1 and 2. Likewise, units are set upright, see equation 3.
- An equation is a statement and ends in a punctuation mark if the sentence structure requires it.
- Special operator symbols are used in the typeset. In particular, some operator symbols known from programming languages are not used, for example for multiplication and exponentiation, see equation 3.

$$\log_2 x = \frac{\ln x}{\ln 2} \quad (1)$$

$$\int_1^x \frac{1}{t} dt = \ln x \quad (2)$$

$$\textit{Cross section} = \textit{width} * \textit{depth} = 42\text{cm} \wedge 2$$



$$A = b \cdot t = 42\text{cm}^2 \quad (3)$$



Please refrain from inserting specific values into a formula. It is sufficient to specify the respective result as a numerical value. The input data should be presented so that they are easy to comprehend, for example in the form of a table.

A common error is the incorrect use of the percentage sign and units. In equation 4, system availability a is formed as the ratio of operating time t_B to total time t_{agg} :

$$a = \frac{t_B}{t_{agg}} * 100 \quad (4)$$



This approach assumes that system availability is always specified as a percentage. Formulae should always be written in such a way that their parameters can be reported in any suitable unit. In equation 5 this is demonstrated by the example of the calculation of a power P from torque M and rotational frequency n .

$$P [kW] = 120 \pi n [rpm] M [Nm]$$



$$P = 2\pi n M. \quad (5)$$



Formulae are numbered consecutively. The number can be omitted for intermediate calculations.

5.9 Algorithms and source code

Software can be used to solve a wide range of tasks. If you create software in the context of your scientific paper, then the underlying algorithms and, if necessary, also source text (or at least excerpts of it) should be included in your paper.

5.9.1 Algorithms

Algorithms are most easily described in so-called pseudocode. As an example, the Euclidean algorithm for determining the greatest common divisor is presented here.

Algorithm 1: Euclid algorithm (pseudo code)

```
1 Euclid(dividend, divisor)
2 as long as b != 0
3     gcd = dividend modulo divisor
4     dividend = divisor
5     divisor = gcd
6 Output dividend
```

To be able to refer to the algorithms, they should each be given a title and number. In addition, a list of algorithms should be provided in the front matter if several algorithms are used. The lines of an algorithm should be numbered so that it is possible to refer to specific passages in the algorithm.

The syntax of the pseudocode is not standardised. However, it should be used consistently within a paper. The essential purpose of pseudocode is to convey intent to a human reader. For this reason, pseudocode is limited to the essentials and usually does not include implementation details.

5.9.2 Source code

If you develop software to solve a task in the course of writing your scientific paper and if this software has special features, you should include an excerpt of the source code that demonstrate its implementation details. However, you should only include the essential passages. The following example demonstrates useful design guidelines:

- To show the indentation, use a monospaced font, i.e. a font in which all characters and letters have the same width;
- It is helpful to use numbered lines so that it is possible to make reference to them;
- Use comments to adequately document the source code;
- The source texts have titles, which are listed in a source text directory in the front matter.

Algorithm 2: Euclid algorithm (Java code)

```

1:  import static javax.swing.JOptionPane.*;
2:
3:  public class Loop {
4:      public static void main( String[] args ) {
5:          String input = showInputDialog( "Please enter first number: " );
6:          int number1 = Integer.parseInt( input );
7:          System.out.print( "The LCD of" + number1 );
8:
9:          input = showInputDialog ( "Please enter second number: " );
10:         int number2 = Integer.parseInt( input );
11:         System.out.print( "and" + number2 );
12:
13:         int lcd;
14:         do {
15:             lcd = number1 % number2;
16:             number1 = number2;
17:             number2 = lcd;
18:         } while ( lcd != 0 );
19:         System.out.print( "is" + number1 );
20:     }
21: }

```

Other formal content such as configuration files or data sets in the form of Extensible Markup Language (XML) or JavaScript Object Notation (JSON) can be documented similarly. Even input/output dialogs can be presented using the guidelines for source code.

5.10 Copies and materials to be handed in

The thesis and, if applicable, its appendices are to be submitted completely and on time to FOM (central examination office). The relevant requirements are documented in the respective framework examination regulations, the subject-specific examination regulations of the respective degree programme and the module description for the final thesis.

5.10.1 Printing and bookbinding

If you submit a printed and bound copy of the thesis, use smooth, white paper with a basis weight of about 80 g/m². The resolution of the print should be at least 600 dpi.

There are three types of binding: spiral binding, glue binding, and thread stitching. Spiral binding is technically simple and usually the cheapest option, however, it has the disadvantage that it can be easily loosened and manipulated. In addition, spiral bound

works are difficult to stack or shelve. Glue binding avoids these disadvantages, is also inexpensive but is not very durable. The most elaborate and technically superior binding is thread stitching. It gives the impression of top quality. However, please keep in mind that the proof copies of theses are usually annotated and afterwards disposed of.

There are also different options to design the cover of the thesis. The simplest is a cover made of plain cardboard. You could also use a transparent front cover. Embossed covers are of higher quality. In any case, it is important to ensure that the front cover shows the title page or that the title page is visible through the transparent cover page.

5.10.2 Digital submission

In addition to the printed copy, the thesis also has to be submitted digitally as a PDF file, which is to be submitted via upload and/or a data storage device. This electronic document is used for archiving purposes. It must therefore have the following properties:

- Fonts and character sets used in it must be embedded in the document,
- The document may not be secured against browsing, printing or extraction,
- It must not be password-protected.

For these reasons, please select a format that is as close as possible to the PDF/A standard¹⁷.

In addition, volatile sources as well as any raw data, source texts, data sheets and so on must be stored electronically (each in its native format) on the data storage device or in an archive that adheres to the ZIP standard, which must then be submitted per upload. The archive should be continuously updated while you work on your thesis. Furthermore, the archive should be structured appropriately using folders with comprehensible titles. This archive may serve as a valuable resource for future research.

¹⁷ Cf. o.V.: (2005), Technical Report.

6 Referencing

In science, knowledge is constantly being gained. It is essential to use these findings for one's own work. In order to do justice to replicability, an essential principle of science, all information that appears in a scientific paper must be traceable to its sources. This requirement is reinforced by normative regulations. In particular, copyright laws and examination laws require that sources are diligently referenced. Negligent or deliberate dereliction of this duty can have considerable legal consequences, even years later, as examples (such as Karl-Theodor zu Guttenberg or Annette Schavan) have shown. Those who do not reference their sources need good nerves. However, those who cite their sources conscientiously and with academic honesty need not fear accusations of plagiarism because of a few careless mistakes.

Sources can be incorporated into a scientific paper in a variety of ways:

- The source can be used unchanged. In the case of texts, this would be a direct quotation. Similarly, figures, tables, source texts and so on can be quoted unchanged. Direct quotations are marked by quotation marks and the respective citation. Direct quotation does not replace the author's own line of reasoning and should be used sparingly. In particular, avoid entire paragraphs compiled purely from direct quotations (collage). Cite the source for unchanged figures and tables.
- The source can be used for inspiration. Thus, an idea, a representation or a fact can be used and paraphrased. For texts, you do not need quotation marks and only reference the source in cases like this. Especially when you use footnotes, such citations are also marked by notations such as *cf.* For illustrations, the reference is supplemented by notes such as *adapted from ...*

You do not need to indicate which figures, tables, or lists you have created on your own. Every source you use has to be referenced; everything else has been created by you.

6.1 Sources

Depending on the topic of the scientific work, the body of source material can be very different. For established topics, there may be a rich body of standard literature, while for new topics, only a small number of sources may be available. For this reason, it is not possible to make a blanket statement about the extent of source to be used. In particular, there cannot be a required minimum of sources. A lack of sources cannot be a reason to refrain from working on a topic. However, if relevant sources are available, they have to be used.

Source materials can have very different features, which have an effect on whether they are quotable and appropriate. Preference should be given to publicly available

primary sources in which facts were originally published by their authors. Secondary sources are also suitable, if possible, in the most recent edition, such as textbooks. It is preferable if these sources are available in the form of books, journals, conference proceedings, speech manuscripts, presentations, lecture videos and so on. The medium in which the source was published is irrelevant. The value of a source is independent of its medium. However, always save ephemeral sources and attach them to the work as a collection of source material.

Problematic as a source is any form of tertiary literature. This includes in particular all encyclopedias (dictionaries, lexicons and generally also Wikipedia). It makes sense to use such sources to familiarise oneself with a subject. However, since the presentation in these encyclopedias can already be considered an interpretation, falsification cannot be ruled out. Therefore, good articles of tertiary literature will refer to the original sources or even to standard literature on the respective topic. You should refer to these sources in your scientific work. Thus, in general, tertiary sources are not appropriate references.



Occasionally, information from sources that are not or only partially public can or should be used. This can be, for example, internal company information or findings by persons who have not published their results. The use of these sources should be vetted very carefully. If they are essential to the scientific paper, their existence and use must be explained in the text. Upon request, readers and especially examiners must be able to gain access to these sources. Under no circumstances may you claim such information as your own sources if it is not the truth.

For some information, the original source is unclear. This is the case for established phrases, for which it is often not clear who used them first. Such a state of affairs can also be explained in the text.

The source of knowledge from general education and knowledge acquired at school is also often unclear. For example, the method of solving quadratic equations is common knowledge. Such generally known facts do not need to be referenced.

6.2 Citation

Correct citation is essential for any scientific work. It enables readers to find the cited sources as well as to gather information about the arguments that do not originate in the author of the paper and it provides new impulses for the reader. Numerous habits of citation have developed depending on the discipline and culture. However, these conventions are very inconsistent. For this reason, many universities and departments establish their own standards, which are more or less obligatory. There is a certain arbitrariness inherent in many of these standards. If the implementation of these standards is demanded down to the last detail, it results in a great deal of effort, which, however, is not met with adequate benefits. For this reason, this chapter is not meant to establish a standard. Rather, its

purpose is to provide a general overview of common methods, as well as a guide to their reasonable application.

In general, a citation should meet the following requirements:

- It should meet the conventions of the field,
- All sources used (text, image, sound, data, software and so on) must be referenced – conversely, a missing source implies that the passage can be attributed to the author of the scientific paper; there is no need to state this explicitly. In particular, the note "own representation" should be refrained from,
- If a citation refers to a sentence, the citation or footnote is placed before the final full stop. If it refers to a whole passage, then it appears after the last full stop of the paragraph,
- Direct quotes should be used sparingly. They should be reproduced in quotation marks or set off and indented in italics so that they can be identified as direct quotes without any doubt,
- If a source is used in an altered form or if it merely serves as inspiration for one's own formulation or design, then this should be indicated by notations such as "cf.", "based on" or "adapted from". These notation are also used when citing by footnote,
- The layout of citations has to be **consistent**.



Roughly two citation styles can be distinguished:

- Footnotes – this style is common in the German-speaking world in the humanities, social sciences, law and economics. In this variant, it is customary to provide extensive information about the source close to the point where it is used; in particular, the page of the reference is always mentioned as well,
- In the text in brackets – this is the established method in the English-speaking world and in Germany in mathematics, computer science, natural sciences and engineering. Here, the information on the source is usually kept rather concise. In specialist literature on mathematics, engineering and the natural sciences, the page of the source is usually not even mentioned because they are easy to find anyway in the usually highly structured sources. In other sciences, the page of the reference should always be included.

Some scientific fields cannot be clearly assigned to these citation habits. For example, one will find both styles in business informatics. However, do not mix the styles within a paper but rather apply one uniformly.

Page numbers are required for papers at FOM irrespective of the style you use.

6.2.1 Citation by footnote

When citing by footnote, the source is typically referenced by stating the author, the year of publication and the page of the reference. If the same source is cited several times in succession, it can be cited in abbreviated form (for example, without the title and year of publication). An example of citation by footnote can be seen in Figure 7

Ein Vorgehensmodell stellt somit „... eine der wesentlichen Grundlagen für einen arbeitsteiligen, einheitlichen, wiederholbaren Prozessablauf“ dar.¹⁰ Ihr Einsatz findet in verschiedensten Fachgebieten statt.¹¹ Chroust

⁷ Vgl. Küffmann(1994), S. 45

⁸ Vgl. Jost(1993), S. 12-13

⁹ Vgl. Pomberger(1996), S. 17

¹⁰ Chroust(1992), S. 37

¹¹ Vgl. Burghardt(1988), vgl. Daenzer(1986), vgl. Schmitt(1991), vgl. Schulte-Zurhausen(1995) und vgl. Sommerville(1992)

Figure 7 : Example of citation by footnote

6.2.2 Citation using brackets

When citing in the running text, the source is placed in brackets (round or square or in the form of slashes). For example, in the frequently used citation style of the Institute of Electrical and Electronical Engineers¹⁸ (IEEE), reference is made to the corresponding number in the bibliography [123]. Alternatively, the reference can be created by an abbreviation that is derived from the author's name, the title of the work, and the year of publication (for example, [KnTAOCP84]). The source is then also listed under this identifier in the list of references.

6.2.3 Citation with short references

Citations with short references include the author's name, the title of the work, the year of publication and, if applicable, the page of the citation. For scientific works published in English, there are established citation standards that can be used for guidance, namely the Harvard referencing style¹⁹ as well as the citation format according to the American Psychological Association²⁰. The exact style of the citation can be modified, possible examples are:

- (cf. Theisen 2017: 24-25) or,
- (see Theisen, 2017, pp. 24-25.).

If multiple works by the same author from the same year are cited, then the different works are marked by small letters, for example (Carraro, 2010a, p.22). Groups of more than three authors are abbreviated in the citation, for example (Carraro et al. 2010). It is important that the source can be identified unambiguously and swiftly in the list of references. The list of references is listed alphabetically by author.

6.3 List of references

The list of references is generally placed after the text and, where applicable, before any appendices and index. However, there is no rule regarding this order, so that you may choose to order them differently.

In general, it is sufficient to create a single list of references. Only in special cases can it be useful to create separate lists for specific sources. In jurisprudential works, for example, it is common to provide a separate index for cited laws.

¹⁸ Cf. IEEE and DIN ISO 960.

¹⁹ Cf. <https://usingsources.fas.harvard.edu>.

²⁰ Cf. <https://www.apastyle.org>.

The entries on the list of references are generally listed alphabetically by the name of the first-named author (exception: see Chapter 6.2.2– IEEE, where the entries are numbered or have systematic identifiers²¹).

Each entry in the list of references should make it easy for a reader to find the source. Thus, as a service to the reader, the bibliographic information should be as complete as possible. Depending on availability, the following information should be included:

- Author(s) or publisher,
- Title (if applicable, also the title of the journal or anthology in which the work appeared),
- Edition (if possible, the current edition should be used if there are not any reasons to use an older edition; the specification of the edition may be omitted for the first edition),
- Place of publication,
- Publisher or institution,
- If applicable, type of publication (dissertation, conference paper, lecture, and so on),
- Publication date,
- ISBN or ISSN,
- DOI if available,
- URL (as accurate as possible) if the source is from the Internet.

The order and typeface of the individual entries should be consistent. An overview of general formats for entries on a list of references can be found in Appendix II.

²¹ For example by a notation in which a cipher is created based on the author's initials and the year of publication.

6.4 Technical aids

Various technical aids can provide substantial support for source work, citations and the compilation of the list of references. It is strongly recommended to make use of these aids. They include systems such as

- BibTeX²²,
- Citavi²³,
- JabRef²⁴,
- Mendeley²⁵,
- Zotero²⁶.

²² <http://www.bibtex.org>.

²³ <https://www.citavi.com/de>.

²⁴ <https://www.jabref.org/>.

²⁵ <https://www.mendeley.com>.

²⁶ <https://www.zotero.org>.

7 Organisational aspects

Within a degree programme, scientific papers are usually submitted as an examination performance, which is intended to demonstrate the student's ability to work scientifically. Even after obtaining the degree, you will produce scientific work to a greater or lesser extent. Many organisational aspects will arise again in the process.

7.1 Examination regulations

The preparation of a thesis as an examination performance requires compliance with the regulations of the central examination office and the examiners. These requirements are subject to possible changes and are of an individual character, especially with regard to the examiners. Therefore, these requirements cannot be detailed here. However, the following is relevant for examination performances:

- The examination regulations are of central importance. It is highly recommended to peruse them,
- Examination regulations are imposed by the examination board or the central examination office. These requirements relate, for example, to:
 - The extent of the material to be submitted (number of copies, data storage devices and so on),
 - Design specifications (for example, permissible bindings),
 - Formalities (declarations, signatures).
- In many cases, examiners define individual requirements, such as the scope of the paper, the number of sources, and so on. If these specifications deviate from the guidelines given here, the examiner's specifications take precedence and are binding.

To avoid difficulties, it is important to strictly adhere to these guidelines. If you are unsure, ask your supervisor or the student advisory service.

7.2 Declaration of academic integrity

In the course of examinations, certain aspects must be observed. First of all, it must be ensured that the paper was actually written by the author and that the authorship is reflected in the paper to a significant degree. To this end, the examination regulations require that the paper has to be accompanied by a legally binding declaration stating that the student has written the paper by themselves and that the student has only used the sources and

aids referenced in the text. Should it turn out – even significantly later – that such a declaration is false, this can have severe legal consequences, for example a fine or the withdrawal of the academic degree.

The exact wording of the statement required is specified by the board of examiners. The board will communicate this wording and it will be binding. The board of examiners will also specify the form and number of declarations to be made and how they are to be signed.

The declaration is not part of the examination performance. Rather, it is an attached document. It is also only of significance in the course of the examination procedure. Technically, it is advisable to provide the declaration as the last page and to sign it in the printed copies. However, the declaration does not have a page number and is not included in the table of contents. If the thesis is to be published later (after the examination has been completed), the declaration should be removed.

Ehrenwörtliche Erklärung

Hiermit versichere ich, dass die vorliegende Arbeit von mir selbstständig und ohne unerlaubte Hilfe angefertigt worden ist, insbesondere, dass ich alle Stellen, die wörtlich oder annähernd wörtlich aus Veröffentlichungen entnommen sind, durch Zitate als solche gekennzeichnet habe. Ich versichere auch, dass die von mir eingereichte schriftliche Version mit der digitalen Version übereinstimmt. Weiterhin erkläre ich, dass die Arbeit in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde/Prüfungsstelle vorgelegen hat. Ich erkläre mich damit einverstanden/nicht einverstanden, dass die Arbeit der Öffentlichkeit zugänglich gemacht wird. Ich erkläre mich damit einverstanden, dass die Digitalversion dieser Arbeit zwecks Plagiatsprüfung auf die Server externer Anbieter hochgeladen werden darf. Die Plagiatsprüfung stellt keine Zurverfügungstellung für die Öffentlichkeit dar.

(Ort, Datum)

(Eigenhändige Unterschrift)

Declaration in lieu of oath

I hereby declare that I produced the submitted paper with no assistance from any other party and without the use of any unauthorised aids and, in particular, that I have marked as quotations all passages which are reproduced verbatim or almost verbatim from publications. Also, I declare that the submitted print version of this thesis is identical with its digital version. Further, I declare that this thesis has never been submitted before to any examination board in either its present form or in any other similar version. I herewith agree/disagree that this thesis may be published. I herewith consent that this thesis may be uploaded to servers of external contractors for the purpose of submitting it to the contractors' plagiarism detection systems. Uploading this thesis for the purpose of submitting it to plagiarism detection systems is not a form of publication.

(location, date)

(genuine signature)

7.3 Blocking notice

Some papers contain confidential information from companies or other institutions. To ensure that such information remains confidential, a paper can be marked with a blocking notice. For this purpose, it is useful to refer to the blocking notice on the title page. For example, the phrase "**Observe the blocking notice!**" can be included on the title page. The blocking notice itself, which then details the precise terms of confidentiality and specifies that the work may only be made accessible to persons who are involved with it in the course of the examination, should immediately follow the title page. Also, the blocking notice is not part of the work, but is an added document. For this reason, the blocking note is not included in the page count and is also not mentioned in the table of contents.

Incidentally, the blocking notice cannot and will not prevent an (automated) plagiarism test. Before applying the blocking notice, it should be verified whether it is really necessary. A blocking notice prevents the author from further exploiting the work and prevents access to the knowledge gained by it. The general scientific value of the work is therefore considerably diminished by a blocking notice. In any case, the supervisor of the paper should agree to a blocking notice.

Sperrvermerk

Die vorliegende Arbeit enthält unternehmensinterne Daten des in der Arbeit genannten Unternehmens. Aus diesem Grund ist die Veröffentlichung der Arbeit nicht erlaubt. Die Arbeit darf außerhalb der Hochschule nur mit der ausdrücklichen Genehmigung des Unternehmens veröffentlicht werden.

(Ort, Datum) (Eigenhändige Unterschrift)

For papers submitted in English:

Blocking notice

The following assignment contains confidential data relating to internal company matters of the company named in the assignment. Therefore, the publication of the thesis is not permitted. Outside the university, the thesis may only be published with the prior written consent of the company.

(Location, date) (genuine signature)

7.4 Sources and appendices

Many papers refer in some form to ephemeral sources, rawmaterial, source texts and other information that is not directly part of the paper itself. Nevertheless, theses sources should be carefully archived and recorded. It is advisable to compile all the material on a suitable data storage device (CD, DVD, USB memory stick). In particular, volatile sources, whose availability is not guaranteed, should be kept in their native formats.²⁷ These volatile sources must be submitted. Additional material may be archived on the data storage device, for example:

- Transcripts or recordings of interviews,
- Raw data from experiments,
- Source code,
- Plans, drawings and sketches.

7.5 Quality assurance

Various mistakes can creep into a paper while you are working on it. They should be eliminated in the course of quality assurance. You should allocate a few days to this phase immediately before printing the paper. During this phase, there should be no more substantial changes to the actual work but only corrections and fine-tuning²⁸. It is relatively exhausting to stay focused while checking one's own text for errors. Many text passages quickly evoke connotations, which can then develop into extensive reflections (Is the issue stated clearly and comprehensively? Is it in the right place?). So, you should be very disciplined in focusing on errors. It can also be helpful to engage someone else for proofreading.

Typical errors that cannot be reliably prevented even by activated spell checkers are, for example:

- Spelling errors (capitalisation, transposition errors),
- Separation errors,



²⁷ No screenshots!

²⁸ Unless you discover serious errors.

- Punctuation errors,
- Text or word duplications,
- Incomplete sentences as a result of cut-and-paste,
- Reference errors (wrong source, wrong figure, wrong quantities and so on).

When the text is effectively finalised, the layout can still be fine-tuned to eliminate minor design flaws. These include:

- Widow lines and orphans (these are single lines of a paragraph at the top or bottom of a page);
- Hyphenation where only one syllable appears in the last line of a paragraph;
- More than three hyphenations in a row;
- Incorrect ligatures (these are merged letters, for example "ff" in "official"; only letters belonging to one syllable are merged, so "rooftop" is correct instead of "rooftop");
- Long tables running over several pages should be divided suitably into extra tables.



It is helpful to look through the text carefully. During this perusal, it is quite easy to ponder which terms should be included in an index. The latter can, therefore, be generated virtually incidentally in the course of quality assurance.



It is not a good idea to postpone the inclusion of source references until the quality assurance phase. If you do this, you have to check almost every sentence against every source. The resulting effort is disproportionately large. Source work should therefore be done continuously during the textual work.



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Appendix

Appendix I: Sample Thesis Title Page



FOM University of Applied Sciences for Economics & Management

University Center [Location]

Bachelor's thesis

in the degree programme [name of the degree programme]

submitted in partial fulfilment of the requirements for the degree

Bachelor of Science (B.Sc.)

on the subject

[Title of the paper]

by

[author's first name last name]

First examiner [Title First name Last name of first referee]

Matriculation number [Author's matriculation number]

Submission date [Date of submission of the thesis]

Appendix II

The entries in the list of references generally have the following format:

Last name author1, first name author1, last name author2, first name author2, & last name author3, first name author3 (year of publication), title of paper, Other publication information.

Journal articles

Author1, first name, author2, first letter first name, author 3, first letter first name (year of publication), title of article, *title of journal*, xx, xxx-xxx.

Example:

Carraro, Carlo, Cian, E., Nicita, L., Massetti, E., de Verdolini, E., (2010), Environmental Policy and Technical Change: A Survey, *International Review of Environmental and Resource Economics* 4 (2), pp. 163-219.

Monographs:

Author1, first name, author2, first letter first name, (year of publication), title of work, edition, place of publication, publisher

Examples:

Shotton, Margaret A, (1989), Computer addiction? A study of computer dependency, London, England, Taylor & Francis.

Kornmeier, M. (2010). Wissenschaftliches Schreiben leicht gemacht für Bachelor, Master und Dissertation, 3rd, updated and expanded edition